NAME: _____ Geometry (FL B.E.S.T.)

4.6 Overlapping Triangles & Special Segments

Triangles have four key special segments. A **median** connects a vertex to the midpoint of the opposite side. An **altitude** is a perpendicular segment from a vertex to the line containing the opposite side (in an obtuse triangle, the altitude from an acute vertex can land outside the triangle). An **angle bisector** splits a vertex angle into two congruent angles. A **perpendicular bisector** is perpendicular to a side at its midpoint, and the **Perpendicular Bisector Theorem** guarantees every point on it is equidistant from the side's two endpoints. The three medians of a triangle always meet at the **centroid**, which divides each median in a $2:1$ ratio from the vertex, and can be found by averaging the three vertices' coordinates. When triangles overlap (sharing a side, an angle, or lying across intersecting lines), a good first step is to redraw them separately, then look for shared parts via the Reflexive Property or the Vertical Angles Theorem. This worksheet combines overlapping-triangle proofs with numeric and coordinate work on medians, altitudes, bisectors, and centroids.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Two overlapping triangles share a side. Which property justifies the side is congruent to itself?

2. Two overlapping triangles share an angle. Which property justifies the angle is congruent to itself?

3. A median connects a vertex to what point of the opposite side?

4. $\overline{AD}$ is a median to $\overline{BC}$. What must be true about BD and DC?

5. An altitude from a vertex is perpendicular to what?

6. An altitude always creates angles measuring what?

7. An angle bisector splits an angle into two angles that are what?

8. What is a helpful first step when triangles overlap in a diagram?

9. Two right triangles share a leg and have congruent hypotenuses. Which postulate applies?

10. In an obtuse triangle, where can an altitude from an acute vertex fall?

11. The centroid divides each median in what ratio, measured from the vertex?

12. A point on the perpendicular bisector of a segment is always equidistant from what?

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13. $\overline{BD}$ bisects $\angle ABC$, with $\angle ABD = (4x - 3)^\circ$ and $\angle DBC = (2x + 11)^\circ$. Find x, each half-angle, and $\angle ABC$.

14. Triangle has vertices $A(0, 8)$, $B(-6, 0)$, $C(6, 0)$. Find the median $\overline{AM}$ from A (to the midpoint of $\overline{BC}$), determine if it is also an altitude, and state what that implies about the triangle.

15. Triangle has vertices $A(3, 9)$, $B(-3, 0)$, $C(9, 0)$. Find the centroid G, the midpoint M of $\overline{BC}$, the median length AM, and verify G divides $\overline{AM}$ in ratio $2:1$ from A.

✓ Answer Key

1. Reflexive Property
2. Reflexive Property
3. The midpoint
4. $BD = DC$
5. The opposite side (or the line containing it)
6. 90°
7. Congruent
8. Redraw the triangles separately
9. HL
10. Outside the triangle, on the extension of the opposite side
11. $2:1$
12. The segment's two endpoints
13. $x = 7$; each half $= 25^\circ$; $\angle ABC = 50^\circ$
14. $AM = 8$; since $AM \perp BC$, it is also an altitude, so $\triangle ABC$ is isosceles ($AB = AC = 10$)
15. $G(3, 3)$; $M(3, 0)$; $AM = 9$; $AG = 6$, $GM = 3$ (ratio $2:1$) ✓